Event-Driven Systems Checklist

A concise operations checklist for Kafka topics, retries, delivery guarantees, and production operations.

1. Topics & Partitioning

- Size partitions for your target consumer parallelism, not just current throughput.
- Choose a partition key that keeps ordering guarantees where they matter (e.g. per-entity-id).
- Avoid hot partitions: watch for skewed keys before they cause consumer lag.
- Plan for partition count growth — you can add partitions, but never safely remove them.

2. Delivery Guarantees

- Decide up front: at-most-once, at-least-once, or exactly-once — each has real cost.
- At-least-once is the default for most brokers; design consumers to be idempotent.
- Exactly-once semantics require transactional producers/consumers and add latency.
- Document the guarantee per topic — don't assume every consumer needs the same one.

3. Producers

- Set acks=all for durability-critical writes; acks=1 trades durability for throughput.
- Batch and compress (lz4/zstd) to cut network and broker load on high-volume topics.
- Handle producer callback errors explicitly — silent drops are the most common data loss bug.
- Use idempotent producers to avoid duplicate writes on retried sends.

4. Consumers

- Commit offsets after successful processing, not before — protects against crash loss.
- Keep consumer group rebalances rare: tune session/heartbeat timeouts for your workload.
- Process messages idempotently; assume redelivery will happen eventually.
- Isolate poison messages with a dead-letter topic instead of blocking the partition.

5. Retries & Error Handling

- Use exponential backoff with jitter for retryable errors; cap total retry duration.
- Separate transient errors (retry) from permanent errors (dead-letter immediately).
- Log the original message and failure reason before routing to a dead-letter topic.
- Alert on dead-letter topic growth rate, not just absolute size.

6. Schema & Compatibility

- Use a schema registry; never ship producer changes without a compatibility check.
- Prefer backward-compatible changes (new optional fields) over breaking ones.
- Version event types explicitly if a breaking change is unavoidable.
- Add contract tests between producers and consumers for critical event types.

7. Monitoring & Operations

- Track consumer lag per partition, not just per topic — averages hide hot spots.
- Alert on under-replicated partitions and offline brokers before they cause outages.
- Monitor end-to-end latency (produce-to-consume), not just broker-side metrics.
- Test broker/AZ failure scenarios in staging before you need them in production.

8. Scaling & Capacity

- Load-test with production-shaped traffic (key distribution, message size) before launch.
- Right-size retention by disk budget and replay needs, not just a default TTL.
- Review consumer group parallelism against partition count — extra consumers idle.
- Revisit partition counts and retention policy on a recurring capacity review.